

DISTANCE Chapter 03

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1 The Illusion of Expansion and the Reinterpretation of the Big Bang

The Universe Is Not Static

The universe never remains perfectly still.

Stars ignite and eventually fade.

Galaxies evolve over billions of years.

Mountains rise and gradually erode.

Atoms exchange energy.

Living organisms grow, reproduce, and die.

Even in situations that appear motionless, countless microscopic processes continue without interruption.

Change is not an occasional feature of the universe.

It is its ordinary condition.

For this reason, humanity has long attempted to explain why the universe changes.

Some explanations have emphasized forces.

Others have emphasized fields.

Still others have described the evolution of physical systems through increasingly sophisticated mathematical formalisms.

Each approach has contributed profoundly to our understanding of specific phenomena.

Yet they often begin from a common assumption.

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They ask how change occurs before asking why a direction of change exists at all.

Consider a simple observation.

A hot object placed beside a colder one does not exchange energy randomly.

The direction is not arbitrary.

Likewise, electrical charge does not move without preference.

Water does not flow equally in every direction.

Chemical reactions do not proceed toward every possible outcome with equal probability.

Again and again, the universe exhibits directionality.

This observation is so familiar that it is often overlooked.

We become accustomed to asking why a particular object moves, while rarely asking why movement possesses a preferred direction in the first place.

DISTANCE begins with this more fundamental question.

Why does the universe exhibit directionality?

The answer proposed here does not begin with motion.

Nor does it begin with force.

Motion is already the visible consequence of something deeper.

Before motion becomes observable, a physical system already possesses a particular state.

That state is characterized not merely by the existence of matter or energy, but by the relationships that remain unresolved among them.

Wherever unresolved differences remain, the system is not complete.

It possesses directionality.

This directionality does not appear later.

It is already embedded within the state itself.

For this reason, DISTANCE distinguishes between two complementary descriptions of the same physical reality.

The first describes the state.

The second describes the directionality associated with that state.

The state is described by Distance.

The directionality is described by Momentum.

Neither concept exists independently of the other.

Neither produces the other.

They simply describe different aspects of the same physical condition.

An unresolved Distance is a state in which equalization remains incomplete.

Momentum is the observable directionality associated with that unresolved Distance.

The distinction may appear subtle, yet it changes the logical foundation of physical interpretation.

Instead of imagining that forces create motion, or that motion itself explains change, DISTANCE begins one level deeper.

Every physical system occupies a particular state of unresolved Distance.

Every such state already possesses directionality.

Observable motion is merely one possible manifestation of that directionality.

From this perspective, equalization is not an event that occasionally occurs within the universe.

It is the ongoing process through which the universe continuously reorganizes unresolved Distance.

Distance describes the present.

Momentum describes its directionality.

Equalization describes the universal process within which both acquire meaning.

Everything that follows in DISTANCE is built upon this single conceptual framework.

Distance and Momentum as Dual Descriptions

If the universe is continuously undergoing equalization, then every physical state can be understood in terms of what remains unresolved.

Some states are nearly equalized.

Others preserve substantial separation.

Some allow immediate exchange.

Others retain separation for extremely long periods.

What matters is not simply that differences exist, but how those differences are present within the current state of the universe.

DISTANCE describes this unresolved condition through two complementary concepts.

The first is Distance.

The second is Momentum.

These concepts should not be understood as separate entities arranged in a sequence.

Distance does not first arise and then produce Momentum.

Momentum does not subsequently act upon Distance from outside.

They are dual descriptions of the same unresolved state.

Distance describes the degree to which equalization remains incomplete.

Momentum describes the observable directionality associated with that unresolved Distance.

One emphasizes state.

The other emphasizes direction.

Consider two bodies of water located at different heights.

If a channel connects them, the state already contains directionality before any visible flow begins.

The higher water does not first possess Distance and then later acquire Momentum.

The unresolved condition and its directionality are already present together.

Once the channel permits exchange, that directionality becomes observable as flow.

Now imagine that the same two bodies of water remain completely isolated by an impermeable barrier.

The unresolved state still remains.

But its directionality does not become observable through immediate exchange.

The distinction does not require us to construct a separate conceptual hierarchy in advance.

It simply shows that the form in which unresolved Distance becomes effective depends upon the conditions of the system.

In ordinary cases, Distance and Momentum can therefore be used according to the emphasis required by the context.

When the purpose is to describe how far equalization remains from completion, Distance is the more useful term.

When the purpose is to describe the direction in which the present state is tending, Momentum is the more useful term.

Neither term replaces the other.

Neither is more fundamental than the other.

They describe the same state from different angles.

This duality appears across every scale.

A temperature difference can be described as an unresolved Distance between thermal states.

The corresponding transfer of energy reveals Momentum.

A pressure difference can be described as Distance.

The directional flow it supports reveals Momentum.

The separation maintained within an atom can be described as Distance.

The directionality through which its internal state continues to reorganize can be described as Momentum.

At larger scales, the same logic applies to planets, stars, galaxies, living systems, and civilizations.

The visible forms change.

The underlying structure does not.

Where equalization remains incomplete, Distance remains.

Where unresolved Distance possesses observable directionality, Momentum is present.

This is why Momentum should not be reduced to the classical expression of mass multiplied by velocity.

That familiar quantity describes one important mechanical form of Momentum.

DISTANCE uses the term more broadly.

Momentum is the observable directionality associated with unresolved Distance.

It may appear as motion.

It may appear as rotation.

It may appear as exchange, reorganization, growth, collapse, or transformation.

Visible movement is only one of its manifestations.

The deeper principle is directionality.

This also explains why equalization should not be imagined as a separate force acting upon the universe.

Equalization is not something added to a preexisting state.

It is the continuing process within which unresolved Distance and observable Momentum acquire meaning.

The universe does not alternate between being static and becoming active.

Even a state that appears motionless may still contain unresolved Distance, internal Momentum, constrained Momentum, or directionality that has not yet become externally observable.

What appears static may therefore be only a locally maintained arrangement within a much larger process of equalization.

At this point, the conceptual structure of DISTANCE can be stated more precisely.

Distance describes the unresolved state of equalization.

Momentum is the observable directionality associated with that unresolved Distance.

Equalization is the continuing process through which the universe reorganizes both.

These are not three stages.

They are not a chain of causes and effects.

They are three perspectives on the same universe.

The next question is therefore not how Distance produces Momentum.

It is how different conditions allow Momentum to become effective, constrained, redirected, or temporarily unobservable within the continuing equalization of the universe.

The Coexistence of Countless Momentum

If Distance describes the unresolved state of the universe, and Momentum describes the observable directionality associated with that state, an important consequence immediately follows.

No physical system can ever be associated with only one Momentum.

Every physical system simultaneously exists within countless unresolved Distance distributed throughout the universe.

Accordingly, it simultaneously participates in countless Momentum.

This is not an exceptional circumstance.

It is the ordinary condition of the universe.

Every physical system exists simultaneously within larger structures while also containing smaller organized structures within itself.

A planet exists within a stellar system.

A stellar system exists within a galaxy.

A living organism exists within its environment while containing innumerable cells, molecules, and atoms.

Every level participates in the ongoing equalization of the universe.

Every level therefore possesses its own associated Momentum.

The observable state of any physical system is consequently never determined by a single Momentum acting in isolation.

Rather, it reflects the coexistence of countless Momentum simultaneously associated with that system across multiple scales.

These Momentum are neither independent nor static.

They continuously influence one another as the universe itself continues its process of equalization.

Their relative significance changes continuously.

Some become increasingly influential.

Others become progressively less significant.

Many remain nearly balanced for extended periods.

The observable directionality of a physical system therefore cannot be understood by examining any individual Momentum in isolation.

It must be understood as the present state arising from their coexistence.

For this reason, Momentum should never be regarded as a permanent property possessed by a physical system.

Nor should it be imagined as something newly created whenever observable change occurs.

What we describe as Momentum is simply the observable directionality associated with the present state of unresolved Distance.

Every observation therefore represents only a snapshot of the continuing equalization of the universe.

As the universe continues to equalize, the coexistence among countless Momentum continuously reorganizes.

The observable directionality changes accordingly.

Yet this changing observable state should never be mistaken for the continual creation or disappearance of Momentum itself.

What changes is the present state of coexistence through which the universe is observed.

This perspective also explains why apparently stable systems remain fundamentally dynamic.

A stable orbit is not the absence of Momentum.

A persistent structure is not free from equalization.

Apparent stability simply indicates that countless Momentum are presently organized in a manner that preserves the observable state.

Such stability is itself one temporary expression of the continuing equalization of the universe.

The universe is therefore never composed of independent objects acted upon by independent Momentum.

It is composed of countless coexisting Momentum whose continuously changing balance determines the observable directionality of every physical system.

Effective Momentum

The existence of Momentum is never the question.

Momentum is always present.

The meaningful question is different.

Among the countless Momentum simultaneously associated with a physical system, which becomes effective under the present state of the universe?

This question should not be understood as asking which Momentum survives while the others disappear.

None disappears.

Every Momentum continues to participate in the ongoing equalization of the universe.

What changes is not their existence.

What changes is their present contribution to the observable state.

Observable directionality therefore never reflects the existence of only one Momentum.

It reflects the current effectiveness of countless coexisting Momentum.

Some contribute strongly.

Others contribute only weakly.

Many remain nearly balanced.

Still others are effectively constrained by the surrounding state of the universe.

The observable Momentum associated with a physical system is therefore always an effective Momentum.

It represents the present directionality emerging from the coexistence of countless Momentum rather than the independent action of any single one.

For this reason, effective Momentum should never be regarded as a permanent characteristic of a physical system.

It is continuously reorganized as equalization proceeds.

Every observation therefore captures only the current expression of the universe.

Another observation made even moments later may reveal a different effective Momentum despite the continued existence of essentially the same underlying Momentum.

Effective Momentum is not a distinct kind of Momentum.

It is simply a description of the observable directionality represented by the present state of countless coexisting Momentum.

It is a description of the observable state produced by the present coexistence of countless Momentum.

This distinction becomes particularly important when interpreting stability.

A system that appears perfectly stable does not lack Momentum.

Rather, the countless Momentum associated with that system presently produce an effective Momentum that preserves the observable state.

Likewise, a rapidly changing system does not suddenly acquire new Momentum.

Its observable evolution reflects the changing effectiveness of Momentum that already coexist within the continuing equalization of the universe.

Effective Momentum therefore bridges the microscopic complexity of the universe and the macroscopic simplicity that we observe.

It allows countless simultaneously existing Momentum to appear as one observable directionality without requiring the universe itself to possess only one Momentum.

Every observable phenomenon is therefore interpreted not as the action of an isolated Momentum, but as the present effective expression of the coexistence of countless Momentum.

Momentum Structures

The concepts developed throughout this chapter naturally lead to another important question.

If countless Momentum continuously coexist throughout the universe, why do we observe recognizable physical structures rather than an indistinguishable continuum of changing Momentum?

Within the DISTANCE framework, this question does not represent a contradiction.

The appearance of observable structures does not interrupt the continuing process of Equalization.

Rather, it is one of the ways in which Equalization continuously proceeds.

Equalization never occurs independently between only two Islands.

Every Island simultaneously participates in countless unresolved Distance extending across multiple scales.

Consequently, every observable state reflects not a single Momentum but the continuously changing coexistence of countless Momentum.

As Equalization proceeds, these Momentum are continuously rearranged under changing local conditions.

The overall direction of Equalization never reverses.

Instead, local Momentum configurations become continuously reoptimized, allowing Equalization to proceed through newly established pathways while preserving its universal

direction.

What appears to us as a physical structure is one temporary consequence of this continuing process.

A Momentum structure should therefore not be understood as a permanent object.

Neither should it be regarded as an independently existing entity.

It is a locally maintained configuration of Momentum whose present arrangement remains distinguishable while Equalization continues.

This distinguishability is neither accidental nor absolute.

A Momentum structure remains observable because its local Momentum configuration continues to maintain an effective boundary with respect to surrounding Momentum structures.

That boundary does not separate it completely from the rest of the universe.

On the contrary, every Momentum structure continuously exchanges influence with countless surrounding Momentum structures while remaining sufficiently distinguishable to be recognized as one coherent structure.

Its identity is therefore functional rather than absolute.

Its boundary is effective rather than permanent.

Its persistence reflects the present state of continuing Equalization rather than its completion.

From this perspective, atoms, molecules, stars, planetary systems, living organisms, and galaxies are not fundamentally different categories of existence.

They are all Momentum structures.

They differ enormously in scale, complexity, and duration, yet each represents the same underlying principle expressed under different conditions.

Within the DISTANCE framework, such locally distinguishable Momentum structures are called Islands.

The word Island should not be understood to imply complete isolation.

Rather, it emphasizes that a Momentum structure possesses an effective boundary allowing it to remain distinguishable while continuously participating in the broader Equalization of the universe.

Every Island therefore exists simultaneously in two complementary ways.

Locally, it maintains its own recognizable Momentum structure.

Globally, it continuously participates in the Equalization of the universe through countless Distance and Momentum extending far beyond its apparent boundary.

What we ordinarily recognize as a physical object is therefore not an independently existing substance.

It is an Island: a locally distinguishable Momentum structure temporarily maintained within the continuing Equalization of the universe.

Redshift as an Observational Signature

Redshift occupies a central place in modern cosmology.

Light received from distant galaxies is observed at longer wavelengths than corresponding light emitted under comparable local conditions.

Within the standard cosmological framework, much of this redshift is interpreted as the stretching of wavelengths caused by the expansion of space.

DISTANCE does not dispute the measured redshift.

It questions whether spatial stretching must be treated as its deepest explanation.

Light does not travel through a perfectly empty, relationship-free container.

Its observable state depends upon the relational environment through which it is generated, propagated, exchanged, and detected.

If the complexity and intensity of momentum-resolution relationships differ between the environment of emission and the environment of observation, then the received signal must reflect that difference.

A photon or wave pattern observed across cosmological scales carries the history of changing relational conditions.

As global energy density decreases and average effective Distance increases, the signal is received within a relational environment different from that in which it was emitted.

What is interpreted geometrically as wavelength stretching may therefore be the observational translation of a deeper decline in relational density, momentum intensity, or energy-exchange effectiveness.

This does not yet constitute a completed quantitative replacement for cosmological redshift theory.

A philosophical reinterpretation is not sufficient by itself.

A valid DISTANCE model must ultimately reproduce the measured relations among redshift, luminosity distance, cosmic age, background radiation, structure formation, and other cosmological observations.

The required normalization between nonlinear relational change and linear wavelength measurement must be mathematically derived.

Yet the conceptual possibility remains important.

Redshift may not prove that an independent substance called space has physically stretched.

It proves that the relationship between distant emission and present observation has changed in a systematic manner.

Modern cosmology describes that change through an expanding metric.

DISTANCE seeks to describe the deeper momentum-resolution process from which such a metric may emerge.

The Internal Observer's Measurement Problem

The observer cannot escape the universe in order to compare its total size against an external ruler.

Every clock belongs to the same changing relational system.

Every unit of length is produced within that system.

Every signal used for measurement participates in its momentum relationships.

This makes cosmological observation fundamentally internal.

The observer resembles the hypothetical intelligence embedded in the spreading ink.

It must infer global change using tools whose own scales change with the relational environment being measured.

This does not make observation impossible.

It makes normalization essential.

Suppose the complexity of momentum-resolution relationships decreases across cosmic development.

The effective spatial scale of rulers, signals, and structures will not remain conceptually independent of that decrease.

Suppose momentum intensity also changes.

The temporal scale provided by clocks and physical processes will likewise vary.

If these changing scales are treated as absolute, the mismatch may appear as the stretching, curvature, acceleration, or deformation of spacetime.

This returns us to the proposition established in Section 1:

When time is viewed through a single scale and space is observed, space appears curved. When space is viewed through a single scale and time is observed, time appears curved.

Cosmic expansion may be another expression of the same observational condition.

If relational change is measured using fixed geometric and temporal assumptions, the nonlinear transformation of density, complexity, and momentum intensity must appear as a changing spacetime geometry.

The observed result remains valid.

What changes is the interpretation of its cause.

Was the Early Universe Truly Small?

The statement that the early universe was smaller may be correct at the observational level.

But the meaning of “small” must be examined.

In conventional cosmology, smallness refers to the scale factor and the geometric separation represented within the model.

In DISTANCE, smallness refers to the compressed observational scale produced by extremely complex momentum-resolution relationships.

When nearly every island participates in intense, overlapping relations, effective Distance is short.

The universe is observed as spatially compact.

As those relations become less dense and less complex, effective Distance increases.

The universe is observed as spatially extended.

Thus the statement “the early universe was small” need not be rejected.

It must be reinterpreted.

The early universe may have been small not because all reality occupied one literal point within an absolute geometry, but because the relational complexity of the universe produced an extremely compressed spatial scale.

Likewise, the present universe may be large not because it has expanded into an external emptiness, but because momentum-resolution relationships have become globally simpler and effective Distances have increased.

The familiar cosmological image is retained at the level of observation.

Its ontological foundation changes.

The Apparent Acceleration of Expansion

Modern observations indicate that cosmic expansion is accelerating.

Within standard cosmology, this result motivates the concept of dark energy or a cosmological constant.

DISTANCE must approach this issue cautiously.

It cannot simply declare that acceleration is an illusion and consider the matter resolved.

The observational data require explanation.

However, if spatial scale depends upon the complexity of momentum-resolution relationships, then a nonlinear decrease in relational complexity may be observed as accelerated expansion.

The rate at which cosmic structures become relationally sparse need not be uniform.

As local concentrations become increasingly isolated, the effective pathways connecting them may decline more rapidly.

A fixed geometric interpretation could translate this changing ratio into an accelerating scale factor.

Similarly, if the intensity of momentum resolution changes independently, the temporal scale used to compare distant events may also shift.

The combined effect could alter the observed relationship among distance, redshift, and cosmic development.

This remains a hypothesis requiring mathematical development.

DISTANCE does not yet possess a complete cosmological model capable of replacing dark energy.

But it identifies a different class of questions.

Instead of asking only what new substance drives spatial acceleration, we may ask:

How does the complexity of cosmic momentum-resolution relationships change?

How does their intensity change?

How do those independent variables alter the temporal and spatial scales through which redshift and distance are observed?

Could part of the apparent acceleration arise from the nonlinear transformation between relational reality and the fixed scales used to measure it?

These questions do not deny modern data.

They challenge the assumption that geometric expansion is the only possible ontological interpretation.

Cosmic Structure Within Global Smoothing

If the universe is smoothing itself, why do galaxies, stars, planets, and other concentrated structures continue to form?

This question appears paradoxical only if smoothing is interpreted as immediate uniformity.

DISTANCE distinguishes local momentum from global momentum.

Global momentum redistributes energy differences across wider relational scales.

It weakens exclusive concentration and increases average energy Distance.

Local momentum organizes differences within temporary islands.

It directs internal relations toward locally optimized equilibrium.

Galaxies and stars are not exceptions to smoothing.

They are local structures created within it.

Matter may converge locally while energy differences are redistributed globally.

A gas cloud collapses into a star because its internal relational conditions generate center-directed local momentum.

At the same time, the star radiates energy and remains embedded within wider global smoothing.

A galaxy forms a coherent island while galaxies as a population become more widely separated.

Local concentration and global dispersion occur together.

The same dual process later appears as gravity and centrifugal behavior, as self-rotation and revolution, and as the persistent tension between island formation and island dissolution.

The universe is neither simply collapsing nor simply expanding.

It is continuously optimizing temporary local structures while reducing imbalance across wider scales.

What appears geometrically as convergence in one relation and expansion in another may therefore arise from the same universal process.

From Spatial Expansion to Distance Maximization

The title of this chapter may appear to deny expansion.

That is not its purpose.

DISTANCE does not deny that the universe is observed as expanding.

It reinterprets what that observation means.

Expansion is real as an observational description.

It may remain indispensable within cosmological calculation.

But the deeper process may be better expressed as distance maximization through energy-gap minimization.

Momentum reduces particular energy gaps.

As those gaps are neutralized, exclusive relationships weaken.

Islands become less tightly bound to specific local structures and more evenly distributed across wider relational networks.

Average energy Distance increases.

This increase is observed geometrically as growing separation.

The universe therefore appears to expand because its privileged energy differences are being progressively neutralized.

The expression “minimize energy gap” and the expression “maximize energy distance” describe the same momentum-driven process.

The first describes what happens to a particular imbalance.

The second describes what happens to the wider relational structure as that imbalance is neutralized.

Cosmic expansion is therefore not the opposite of equalization.

It is one of equalization’s observable forms.

The Universe Is Mixing Itself

The history of the universe need not be understood primarily as the history of a geometric container increasing in size.

It may instead be understood as the history of a relational structure redistributing its differences.

The early observable universe was extraordinarily dense.

Its momentum-resolution relationships were complex and intense.

It was therefore observed through a compressed spatial scale and a rapid temporal scale.

As global smoothing proceeded, average density decreased.

Relational complexity became lower across cosmic scales.

Effective Distances increased.

The universe came to appear larger.

Within that global process, local islands continued to form.

Galaxies rotated.

Stars condensed.

Planets emerged.

Life organized itself.

Every local structure temporarily resisted complete equalization while remaining part of the larger smoothing process.

The universe is therefore not merely expanding.

Nor is it merely dispersing.

It is continuously mixing itself.

It creates temporary concentrations while weakening absolute separation.

It generates local order while reducing global imbalance.

It shortens effective Distance where energy gaps become active momentum and increases average Distance as those gaps are neutralized.

What modern cosmology observes as expansion may be the geometric image cast by that deeper relational transformation.

The balloon is not necessarily wrong.

It is incomplete.

It shows the dots moving apart.

The ink reveals why they may appear to do so.

Beneath both lies the same process:

energy gaps becoming momentum,

momentum reorganizing relationships,

and the universe moving continuously toward wider smoothing.

The universe may still be described as expanding.

But expansion is not necessarily its deepest action.

Its deeper action is equalization.

The universe is not simply growing larger. It is becoming relationally thinner as it continually smooths itself.